

Quiz (Habanero)

- Q1.** A research team measures 2.5 inches of rainfall in a forest and 1.8 inches in a nearby meadow. What is the ratio of forest rainfall to meadow rainfall?
- (A) 1.35 : 1
(B) 1.39 : 1
(C) 1.45 : 1
(D) 1.51 : 1
(E) 1.58 : 1
- Q2.** If 15 gallons of water are used to irrigate 30 square feet of a garden, how many gallons are needed for 45 square feet?
- (A) 10 gallons
(B) 15 gallons
(C) 22.5 gallons
(D) 25 gallons
(E) 30 gallons
- Q3.** In an ecosystem, the ratio of predator to prey is 1 : 6. If there are 36 prey, how many predators are there?
- (A) 5
(B) 6
(C) 8
(D) 10
(E) 12
- Q4.** A water tank can hold 2400 gallons of water. If 300 gallons of water are already in the tank, what fraction of the tank is filled?
- (A) $\frac{1}{8}$
(A) $\frac{1}{6}$
(A) $\frac{1}{5}$
(A) $\frac{1}{4}$
- Q5.** A temperature increase of 4°C causes a 2-inch rise in sea level. If the temperature increases by 12°C , what is the rise in sea level?
- (A) 5 inches
(B) 6 inches
(C) 8 inches
(D) 10 inches
(E) 12 inches
- Q6.** A park has a ratio of trees to benches of 12 : 1. If there are 48 trees, how many benches are there?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q7.** A cyclist travels 25 miles in 2 hours. If she maintains the same rate, how many miles will she travel in 4.5 hours?
- (A) 50 miles
(B) 56.25 miles
(C) 62.5 miles
(D) 67.5 miles
(E) 70 miles
- Q8.** The ratio of sunny days to rainy days in a month is 5 : 3. If there are 20 rainy days, how many sunny days are there?
- (A) 25
(B) 30
(C) 31
(D) 32
(E) 33

Open Ended Questions (FRQ)

- Q9.** A city has a ratio of parks to residents of 1 : 1000. If the city currently has 25 parks and a population of 35,000, how many more parks should be built to maintain the current ratio?
- Q10.** A farmer uses 15 gallons of water to irrigate 100 square feet of land. If she wants to irrigate 500 square feet of land, how many gallons of water will she need? Show your work and explain your answer.

Chef's Tips

- Read each question carefully before answering
- Use the given information to solve the problem
- Check your work and explain your reasoning